From Show and Tell to Attention and Transformers: An Empirical Study of Architecture for Image Captioning

Hao Yang
Georgia Institute of Technology
hphd9@gatech.edu

Qun Wei
Georgia Institute of Technology
qwei49@gatech.edu

Oscar A. Martinez
Georgia Institute of Technology
omartinez33@gatech.edu

Yongjian Wei
Georgia Institute of Technology
ywei364@gatech.edu

Abstract

Image captioning requires effective integration of visual understanding and natural language generation. This work proposes a unified image captioning framework for a controlled comparison of key design choices, focusing on visual representations, decoding architectures, and embedding strategies. A pretrained ResNet-50 [4] CNN encoder extracts global and spatial image features, which are combined with LSTM, attention-based LSTM, and Transformer decoders to form four model variants. The classical Show and Tell [14] model is used as the baseline, while the other variants introduce spatial attention [15] and Transformer-based decoding [12]. In addition, different word embedding strategies are evaluated, including randomly initialized embeddings and pretrained GloVe [11] embeddings (frozen and fine-tuned). Experiments on the Flickr8K [5] dataset using standard metrics show that spatial features improve visual grounding, while Transformer decoders further enhance performance by better modeling contextual dependencies. Pretrained embeddings provide only marginal gains in low-data settings, with limited benefit from fine-tuning. Overall, the results suggest that performance improvements are primarily driven by stronger representation learning and attention mechanisms, while accurate visual grounding in complex scenes remains a key challenge.

1. Introduction / Background / Motivation

1.1. Project Objective: Automatic Image Description

The goal of this project is to build a system that can look at an image and automatically generate a natural, human-like sentence describing what is happening in it. Beyond simply generating captions, the main objective is to systematically study how different design choices affect caption quality.

1.2. Current Practice and Key Limitations

Modern image captioning is typically formulated as encoder-decoder, with a CNN extracting visual features and a sequence model generating the description. Show and Tell [14] passes a single global vector to an LSTM; Show, Attend and Tell [15] adds soft attention over spatial feature maps; and more recent work replaces recurrent generation with Transformer self- and cross-attention [12]. Prior work often introduces several of these changes at once, so it remains unclear which component contributes most to performance, which motivates a more systematic, controlled comparison.

1.3. Impact and Motivation

Solving this problem matters because countless images online lack useful text descriptions. A successful automated model greatly improves internet accessibility for visually impaired users, enhances image search capabilities, and helps organize large visual datasets. With wearable devices like smart glasses and mobile assistive technologies becoming more common, image captioning can also support real-time scene understanding by describing users' surroundings. A clearer understanding of which model components actually drive performance can guide the design of more efficient and robust captioning systems for these practical deployments.

1.4. Dataset Details

Following the *Datasheets for Datasets* framework [3], we provide additional context on the Flickr8K dataset [5]. Flickr8K was originally introduced by Hodosh et al. for sentence-based image description and ranking, where the original task was caption retrieval from a candidate pool rather than caption generation. The dataset contains 8,091 images sourced from Flickr, each paired with five independent captions written in English by Amazon Mechanical Turk workers, giving roughly 40,000 image-caption pairs in total. We use the standard 6,000 / 1,000 / 1,000 train/validation/test split, treating each image-caption pair as a separate training example (yielding 30,000 training pairs) and computing evaluation metrics against all five reference captions per image. Compared to other commonly used image captioning benchmarks, Flickr8K sits on the smaller end of the spectrum [6], with Flickr30K roughly $4\times$ larger and MS COCO Captions roughly $15\times$ larger.

Notably, the images were deliberately curated to focus on scenes involving people or animals performing some

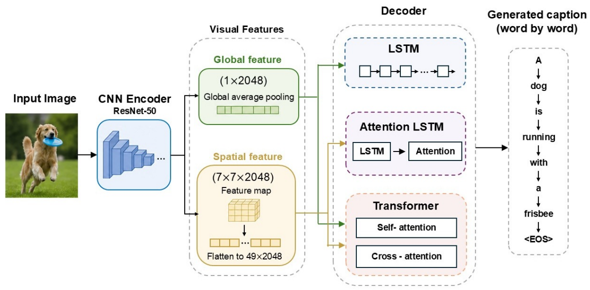


Figure 1. Overview of the unified image captioning framework with interchangeable encoder and decoder components.

Model	Encoder	Decoder	Description
M1	Global	LSTM	Show and Tell baseline
M2	Spatial	Attn-LSTM	Attention-based LSTM
M3	Spatial	Transformer	Transformer w/ spatial
M4	Global	Transformer	Transformer w/ global

Table 1. Model Configurations.

action, which biases the topic distribution toward people, dogs, and outdoor activities and away from static scenes or domain-specific imagery like medical or aerial photography. The English-only captions and the predominantly US-based Mechanical Turk worker pool also mean that the linguistic patterns we learn are unlikely to generalize to other languages or cultural framings of the same images. Hodosh et al. explicitly positioned Flickr8K as a research benchmark for sentence-based image description rather than a production-ready corpus, and we treat it the same way. Despite these limitations, Flickr8K is a deliberate fit for this study, as the dataset is small enough to make iterative hyperparameter tuning across four architectures feasible on modest compute, while still being large enough to expose meaningful differences between architectures.

2. Approach

2.1. Overall Framework

Figure 1 illustrates the unified image captioning framework, which follows a standard encoder-decoder paradigm. A pretrained ResNet-50 [4] is used as the shared visual encoder to extract image representations, while a sequence decoder generates captions in an autoregressive, word-by-word manner. From the encoder, two types of visual features are derived: a global feature obtained via global average pooling, and spatial features extracted from the final convolutional feature map that preserve region-level information. On the decoder side, we consider three architectures: LSTM, attention-based LSTM, and Transformer. By combining different visual representations with different decoders, we construct four model variants (M1 to M4), as summarized in Table 1. The global feature + LSTM model (M1) corresponds to the Show and Tell [14] framework and serves as our baseline, while the other variants introduce spatial attention or Transformer-based decoding. By isolating these components against a fixed training and evaluation pipeline, the framework aims to identify which factors contribute most to caption quality, rather than confounding architecture changes with data or training-setup differences.

2.2. Visual encoding

We use a pretrained ResNet-50 [4] as the shared visual encoder for all models. Show-and-Tell [14] originally used an Inception-based encoder; we adopt ResNet-50 for its stronger representations, broad adoption, and modular compatibility across model variants. ImageNet [2] pretrained weights are used with the backbone frozen during training.

Two visual representations are derived from the same backbone. **Spatial encoding** uses the output of the final convolutional block before global average pooling, producing a $7 \times 7 \times 2048$ feature map that is reshaped into 49 region features and projected to $(49, 512)$ for attention-based decoding. **Global encoding** is derived from the same spatial representation by average-pooling the 49 projected region features into a single 512-D image vector. The two representations capture complementary information: global features encode holistic scene semantics, while spatial features preserve region-level information for object-word alignment. Spatial features are expected to benefit caption quality through finer visual grounding, whereas global features provide a simpler, strong baseline.

2.3. Caption decoding architectures

All models formulate caption generation as autoregressive next-token prediction conditioned on image features and previously generated words. Three decoder families are compared:

- **LSTM (M1)** follows the encoder-decoder formulation of Show and Tell [14]: a global image feature initializes the recurrent decoder's hidden state, and tokens are generated step by step.
- **Attention-LSTM (M2)** extends the recurrent decoder with soft visual attention [15] over the 49 spatial regions at each step, enabling dynamic region selection during generation.
- **Transformer (M3, M4)** replaces recurrence with stacked masked self-attention and cross-attention layers [12]. Caption tokens attend both to previous words (causal self-attention) and to encoded visual regions (cross-attention). M3 uses spatial visual features; M4 uses the same global feature as M1, isolating the effect of decoder family under a fixed visual representation.

These architectures span increasing modeling capacity, from sequential recurrence, to recurrence with attention, to fully attention-based decoding, enabling a controlled comparison of decoder design under shared training and evaluation settings.

2.4. Word Embeddings

Three word embedding strategies are considered: **random initialization**, **pretrained GloVe [11] embeddings with frozen weights**, and **pretrained GloVe embeddings with fine-tuning**. In the first setting, embeddings are learned from scratch during training, serving as a baseline without external linguistic knowledge. In the second setting, embeddings are initialized using pretrained GloVe vectors (trained on large-scale text corpora to capture global word co-occurrence statistics) and kept fixed, providing stable semantic representations while reducing overfitting. In the third setting, pretrained embeddings are further fine-tuned, allowing adaptation to the captioning task and potentially improving alignment between visual and textual features.

The pretrained GloVe embeddings are 300-dimensional, while the model embedding size is 512. To address this mismatch, a linear projection layer is applied to map the embeddings into the model space. To ensure fair comparison, all models share the same vocabulary and embedding dimensionality, and only the embedding initialization and trainability are varied. Pretrained embeddings are expected to provide improvements over random initialization, while the additional benefit of fine-tuning may be limited due to the limited size of the training data.

2.5. Evaluation Metrics

Caption quality is evaluated using four standard metrics: **BLEU** [10], **METEOR** [1], **ROUGE**-L [8], and **CIDEr** [13], which are widely adopted in prior image captioning studies. BLEU measures n-gram precision between generated captions and reference captions, emphasizing local word overlap. METEOR incorporates both precision and recall with additional alignment mechanisms such as stemming and synonym matching, making it more sensitive to semantic similarity. ROUGE-L is based on the longest common subsequence and captures sentence-level structure and fluency. CIDEr measures consensus between generated captions and multiple references using TF-IDF weighting, emphasizing the importance of informative and distinctive words.

Among these metrics, CIDEr is considered the primary metric in this work, as it is specifically designed for image captioning and better reflects human judgment by accounting for consensus across multiple references. BLEU and ROUGE-L provide complementary insights into lexical overlap and structural similarity, while METEOR offers additional sensitivity to semantic variations. All metrics are reported to provide a comprehensive evaluation of model performance.

3. Experiments and Results

3.1. Experimental setup and model configurations

Code repositories and prior sources. The full implementation is available at our GitHub repository. The training pipeline, encoders, attention module, Transformer decoder layer, and evaluation utilities were written from scratch in PyTorch; we referenced patterns from our own prior CS 7643 coursework (Assignments 2 and 3) for scaffolding such as training-loop structure and evaluation harnesses, but no code was copied verbatim. Third-party dependencies are limited to `torchvision`'s ImageNet-pretrained ResNet-50 weights, the standard NLTK BLEU/METEOR implementations, and a CIDEr scorer adapted from the public coco-caption toolkit. This report is built from the course-provided LaTeX template.

Dataset. Flickr8k [5] (partition into training-6k, dev-1k, and test-1k data).

Loss function. All models were trained to minimize the cross-entropy loss, which calculates the per-token probability error of the generated caption against the ground truth.

Optimizer. The AdamW [9] optimizer was preferred over standard Adam. AdamW decouples weight decay from the gradient update step, which is recommended for training complex sequence models like transformer and LSTM, leading to better generalization.

Model	LR	WD	Hid.	Lyrs.	Heads	Drop.
M1	1e-4	0.05	512	2	NA	0.1
M2	1e-4	0.05	512	1	1	0.1
M3	1e-4	0.05	512	4	8	0.3
M4	1e-4	0.05	512	4	8	0.3

Table 2. Model-specific best hyperparameters. WD: weight decay; Hid.: hidden/dim; Lyrs.: layers; Drop.: dropout. All models are trained for 10 epochs with batch size 16.

Transfer Learning. Due to smaller dataset (Flickr8k), transfer learning was utilized to accelerate convergence across all the models. The weights of the Resnet-50 encoder were strictly frozen during training. The model primarily focuses on optimizing the decoder's ability to map established visual representations to text.

Hyperparameter Tuning Strategy. The hyperparameters were refined through iterative tuning, with specific attention to regularization mainly due to relatively small size of Flickr8k dataset use (Table 2):

- **Batch size:** A batch size of 16 was utilized. During tuning, smaller batch size provided a regularization effect through slightly noisier gradient updates, which helped the models escape sharp local minima.
- **Learning Rate Scheduler:** A learning rate scheduler was implemented, utilizing a warmup phase (first 2 epochs) followed by annealing. The warmup phase was to prevent early divergence, especially for transformer decoder, while the annealing allowed the optimizer to smoothly settle into the final optimal weights in the last few epochs.
- **Regularization:** to combat overfitting on the small dataset, a dropout rate ranging from 0.1 to 0.3 were applied to the encoder and decoder models respectively. And a weight decay of 0.05 was enforced via the AdamW optimizer to penalize overly large network weights.

Problems anticipated and encountered. We anticipated that the Transformer decoders would be the highest-risk component, since fitting an attention model on ∼6K training images is ripe for overfitting. Several of our up-front choices reflected that assumption: freezing the ResNet-50 backbone to avoid catastrophic forgetting, keeping the decoder narrow at hidden size 512, and sharing a single small vocabulary across all four models so that the architecture comparison would not be confounded by tokenization.

Despite these precautions, the first runs did not work as intended. Looking at the M3 loss curves, training and validation loss diverged relatively early. We worked through this iteratively rather than in one shot: each of the regularization choices already described (AdamW with decoupled weight decay, per-model dropout in $\{0.1, 0.2, 0.3\}$, and a small batch size for added gradient noise) were selected to address the divergence. At inference time, switching from greedy decoding to beam search produced an addi-

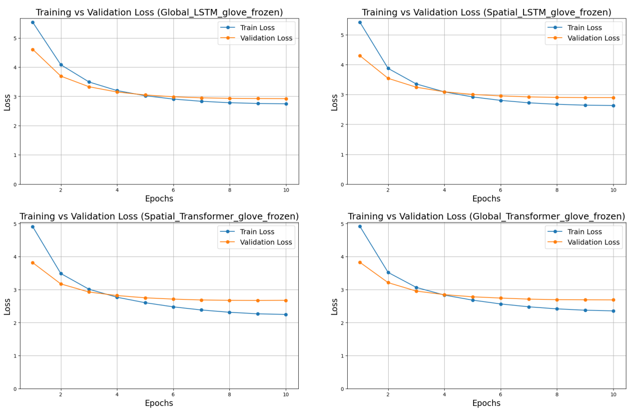


Figure 2. Training convergence comparison between M1 (top left), M2 (top right), M3 (bottom left), and M4 (bottom right).

tional gain of up to 3% across our reported metrics. None of these were the first thing we tried.

Training results. Figure 2 illustrates the loss curves for M1 through M4 across 10 epochs. The LSTM-based models (M1 and M2) demonstrate relatively stable convergence, with training and validation loss aligning closely and plateauing around 2.9. Conversely, despite aggressive regularization, the Transformer models (M3 and M4) exhibit a noticeable divergence between training and validation loss starting around epoch 4: training loss continues to decline while validation loss plateaus, a trend typical for high-capacity models on small datasets. Despite this divergence, the Transformer models ultimately reach a lower final validation loss (around 2.65 to 2.7) than the LSTM baseline.

3.2. Architecture Comparison

This section evaluates the four model variants to analyze how specific architectural choices influence caption quality. To ensure optimal sequence generation, all evaluation metrics were calculated using beam search decoding strategy. Table 3 summarizes the quantitative results for each model.

A consistent observation across the result is the impact of Visual Representation: Global vs Spatial (M1 vs. M2 & M4 vs. M3): preserving spatial features significantly improves performance for both LSTM and transformer based decoders.

- (M1 vs. M2): the baseline model forces the entire image into a single global vector, discarding localized visual details. By transitioning to CNN spatial features paired with an Attention-LSTM (M2), the decoder has the ability to focus on specific image regions for each generated word. This localized visual grounding generates higher caption accuracy of CIDEr score from 0.37 to 0.42.
- (M3 vs. M4): the transformer decoder shows a similar reliance on spatial information. The global feature discards the visual region details that cross-attention mechanisms in transformer decoder are designed to exploit. It results in more generalized and less accurate captions. On the other hand, with spatial feature map provided, the transformer can align distinct image regions with visual tokens, allowing the generated text to be tightly grounded to specific objects and backgrounds. As a result, M3 pushes the CIDEr score to a project-high of 0.47.

Model	B-1	B-2	B-3	B-4	MET.	R-L	CIDEr
M1	0.56	0.37	0.24	0.15	0.36	0.44	0.37
M2	0.58	0.40	0.26	0.17	0.37	0.46	0.42
M3	0.58	0.40	0.27	0.18	0.38	0.46	0.47
M4	0.57	0.39	0.25	0.17	0.37	0.45	0.43

Table 3. Quantitative architecture comparison (best hyperparameter model). B-n: BLEU-n, MET.: METEOR, R-L: ROUGE-L.

For the impact of decoder architecture, M3 outperforms M2 because the Transformer decoder uses the same spatial image features more effectively than the Attention-LSTM decoder. In M2, caption generation depends on a recurrent hidden state, so visual and language context must be carried forward step by step. This can limit how well the decoder preserves earlier information, especially when generating longer phrases. In contrast, M3 uses masked self-attention, allowing each word position to directly reference previous caption tokens, while cross-attention links those tokens to spatial image regions. This structure gives M3 a more flexible way to model object relationships, phrase structure, and image-text alignment. Therefore, with the same spatial encoder input, M3 produces more coherent and better-grounded captions than M2.

3.3. Embedding Comparison

To assess the impact of prior semantic knowledge on the image caption generation process, three models, baseline (M1: Global+LSTM), M2 (Spatial+Attention-LSTM) and M3 (Spatial+Transformer), were evaluated across three word embedding configurations. Although the experimental results in Table 4 demonstrate that the introduction of pre-trained GloVe embeddings yield negligible improvements across all architectures, the underlying reasons for this lack of impact could differ.

Model	Embedding	B-1	B-2	B-3	B-4	MET.	R-L	CIDEr
M1	Random	.568	.379	.245	.155	.355	.445	.370
	GloVe frozen	.562	.372	.241	.152	.356	.442	.373
	GloVe finetune	.558	.367	.237	.155	.355	.445	.370
M2	Random	.546	.369	.243	.154	.368	.454	.417
	GloVe frozen	.572	.386	.255	.165	.368	.452	.421
	GloVe finetune	.573	.389	.260	.169	.370	.455	.422
M3	Random	.581	.398	.271	.181	.383	.465	.473
	GloVe frozen	.582	.398	.271	.183	.379	.463	.467
	GloVe finetune	.587	.404	.275	.184	.380	.463	.478

Table 4. Embedding ablation across M1 (Global+LSTM), M2 (Spatial+Attn-LSTM), and M3 (Spatial+Transformer) under three word-embedding configurations.

For baseline M1 and M2 architecture, the primary limitation is architecture rather than semantic. M1 compresses the visual input into a single global vector and the LSTM decoder lacks the visual context to effectively align with the rich, pre-existing semantic space provided by GloVe. M2 introduces spatial feature and attention mechanism, the performance from random to GloVe embeddings shows improvements in BLEU and remain marginal in the other metrics. Conversely, for the M3 architecture, the negligible difference between different embeddings indicates a limitation of data scale. Because the training set is relatively small, the transformer reaches the performance ceiling and begins to overfit as illustrated in the earlier training results. Therefore, it does not fully exploit the high-dimensional semantic spaces provided by the GloVe embeddings, and embeddings don't play a decisive factor in the results.

3.4. Qualitative Analysis

To further understand model behavior beyond quantitative metrics, we present qualitative comparisons across three levels of difficulty, as shown in Figure 3. Each column (easy, medium, challenging) contains two representative examples, allowing us to assess consistency within each difficulty level.

In easy cases (Figure 3, left), the main subject and action are visually salient against clean, simple backgrounds, with the subject occupying a large portion of the image. Under these conditions, all four models generate nearly identical and accurate captions, suggesting that when foreground and context are well-separated, even simpler architectures reliably capture the dominant global semantics and architectural differences have minimal impact on performance.

In medium cases (Figure 3, middle), the primary subject remains clear, but the scene becomes more complex with cluttered backgrounds, partial occlusion, and less salient secondary objects such as a tennis ball or a handheld wand. The models begin to diverge: M1 captures only the dominant activity (e.g., "playing in the water"), M2 picks up the secondary object but omits scene context, and M3 and M4 integrate both, with M4 occasionally adding attribute-level details such as color. The same general progression holds across both medium examples, though M3 and M4 occasionally drift into mild hallucination (e.g., M3's "swinging on her head"), foreshadowing the instability we see in the challenging column.

In challenging cases (Figure 3, right), the entire scene becomes visually complex, with multiple objects, cluttered layouts, low contrast between foreground and background, and at times poor lighting. All four models fail to fully capture the ground-truth semantics but exhibit distinct error patterns: M1 and M4 produce plausible but generic captions, M2 partially improves environmental understanding (e.g., "lake"), while M3 generates an unrelated scene (e.g., involving a building), indicating hallucination. This suggests that when visual evidence is weak or ambiguous, stronger generative capacity can lead to increased instability rather than to better grounding.

Overall, model differences become more pronounced as scene complexity increases. Attention-based models improve fine-grained detail capture in moderately complex scenes, but robust visual grounding under clutter, occlusion, and low-visibility conditions remains a key challenge for all four architectures, particularly when trained on a relatively small dataset like Flickr8K.

4. Discussion and Conclusion

Our controlled experiments with four different models reveals a fundamental tradeoff between complexity and representation power. The baseline M1 (Global+LSTM) is computationally efficient but lacks complexity and capacity, resulting in worst performance in the comparison. M2 adds complexity by using attention mechanisms to align text tokens with image local regions, and the performance sits in the middle ground. M3 (Spatial+Transformer) achieves the highest metrics but easy to overfit with small training data. M4 underperformed compared to M3 due to inefficient use of cross-attention with compressed global visual feature.

Looking back at the project's stated goal of isolating which architectural axis matters most for caption quality, the controlled comparison succeeded: it produced a clear ranking (M3 $>$ M4 $\approx$ M2 $>$ M1) and identified spatial features as the dominant axis under our fixed training and evaluation setup. The project fell short on generalization to challenging scenes, however, where all four models struggle with small-object and interaction cases. We attribute that gap primarily to Flickr8K's limited scale and topic skew, both of which constrain how robustly any of these architectures can fit the long tail of caption variation.

The results clearly demonstrate that preserving spatial visual features is critical for accurate image-text alignment. Furthermore, with rich spatial features from the encoder, the Transformer architecture outperforms the attention-LSTM approach by utilizing masked self-attention and cross-attention to generate highly contextual captions for image.

The architecture mirrors the task: a frozen ResNet-50 encoder maps pixels to features, and a learned sequence decoder (LSTM or Transformer self/cross-attention, alongside the word embeddings and output projection) maps those features to language under token-level cross-entropy. The pipeline is implemented in PyTorch with standard preprocessing (ImageNet image normalization, lowercased whitespace-tokenized captions) and beam-search inference (beam width 3, max length 50).

The most direct extension is scaling: Flickr30K and eventually MS COCO Captions, since M3's overfitting on Flickr8K is the cleanest evidence we are limited by data rather than by architecture. On the architecture side, a fully attention-based pipeline (ViT + Transformer decoder, alongside a contemporary baseline like BLIP-2 [7]) extrapolates the comparison from M1 to M4, and CIDEr-rewarded reinforcement learning is a complementary objective-level direction since our primary metric is non-differentiable.

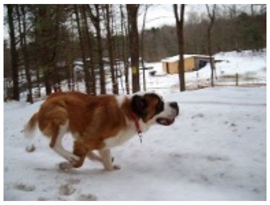
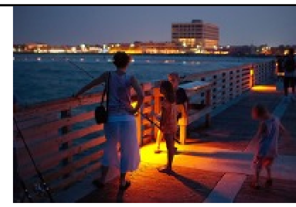
	Easy Case	**Medium Case**	**Challenging Case**
Image			
GT (5)	**(1)** A beautiful brown and white St Bernard running in the snow. **(2)** A St Bernard lunges through the snow. **(3)** Large St Bernard dog wearing red collar galloping through the snow. **(4)** St Bernard dog running in the snowy field. **(5)** The large brown and white dog is running through the snow .	**(1)** A brown dog is running through neck-deep water carrying a tennis ball. **(2)** A brown dog splashes in the water while carrying a ball in its mouth. **(3)** A brown dog swims through water outdoors with a tennis ball in its mouth. **(4)** A red dog, holding a ball, splashes through vegetation filled water. **(5)** Brown dog with tennis ball in mouth, in water and bushes .	**(1)** A child and a woman are at waters edge in a big city. **(2)** a large lake with a lone duck swimming in it with several people around the edge of it. **(3)** A little boy at a lake watching a duck. **(4)** A young boy waves his hand at the duck in the water surrounded by a green park. **(5)** Two people are at the edge of a lake, facing the water and the city skyline .
M1	a brown and white dog is running through the snow.	a brown dog is playing in the water	two people are standing on a beach
M2	a brown and white dog is running through the snow.	a brown dog is playing with a tennis ball	three people are standing in the water near a lake
M3	a brown and white dog is running through the snow.	a brown dog is playing with a tennis ball in the water	a man in a black shirt is standing in front of a building
M4	a brown and white dog is running through the snow.	a brown and white dog is playing with a tennis ball in the water	two people are standing in front of a large body of water
Image			
GT (5)	**(1)** Two brown dogs are playing with each other in the snow . **(2)** Two brown dogs chase each other in the snow . **(3)** Two brown dogs play in the snow . **(4)** Two dogs playing in the snow . **(5)** Two dogs playing or fighting in the snow .	**(1)** A girl in a ballet fairy costume is touched by a wand. **(2)** A young girl performing the nutcracker. **(3)** The little girl is standing on a stage dressed up as a pink fairy. **(4)** The tiny girl dressed in a butterfly ballet costume , waves her wand near a nutcracker statue. **(5)** Young girl waving turquiose leotard and pink tutu with wings waving wand .	**(1)** A mother and children is fishing on a boardwalk at night. **(2)** A woman and three children stand on a dock with a fishing pole. **(3)** A woman stands with children on a boardwalk at night overlooking the sea. **(4)** Some people on a pier at night with one girl fishing off it. **(5)** Woman with three children fishing over boardwalk in the evening.
M1	two brown dogs play in the snow.	a girl in a pink shirt is playing with a <unk>	two people are sitting on a beach
M2	two brown dogs are playing in the snow	a young girl in a pink dress is playing with a pink toy	two people stand on a dock at the water
M3	two brown dogs play in the snow	a little girl in a pink dress is swinging on her head	two people are standing on a dock looking at the water
M4	two brown dogs play in the snow	a little girl in a pink dress is playing with a hula hoops	a man and a woman are sitting on a bench at the beach

Figure 3. Qualitative caption comparison across three difficulty levels (easy, medium, challenging) for the four model variants. “GT (5)” lists the five reference captions per image.

Work Division

Work was distributed so that every member contributed to both implementation and writing. Model variants were primarily owned as follows: Hao led M3, Yongjian led M2, and Oscar led the initial M1 and M4 implementations; Qun and Yongjian jointly developed the LSTM attention decoder shared across variants. Cross-cutting work (the unified codebase, embedding ablation, evaluation harness, and LaTeX transcription) and individual report sections were divided across the team as detailed in Table 5.

Members	Contributions
Hao Yang	Developed the implementation of the M3 model, beam search decoding, and evaluation metrics. Contributed to the design of the overall codebase structure and the unification of all four models (M1 to M4). Ran experiments for the M3 model. Initiated report organization and completed Abstract, Section 2, and Section 3.4.
Yongjian Wei	M2: Spatial encoder and LSTM attention code development. Run experiments for M2 model. Report writing section 1, section 3.2.
Qun Wei	Co-development of multi-layers LSTM attention decoder with Yongjian. Implemented Global encoder and a unified model for M1, M2, M3, and M4 for team collaboration. Run experiments for embeddings for all models. Report writing section 3.1, 3.3 and 4.
Oscar A. Martinez	Initial M1 and M4 code development (later superseded by the team's unified `Model_IC`). Creation of Optuna scripts for hyperparameter tuning (M1 and M4). Transcription of report from Word to LaTeX, and overall report formatting.

Table 5. Contributions of team members.

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